PETW Step 3 Extreme-Value Analysis of Thunderstorm Winds and Mixed-Wind Climates: Wind-Speed Standardization, Mechanism Classification, and Uncertainty Propagation

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Scientific status: `not_scientifically_approved`; `candidate_evidence_only` for any future PETW design use. This literature review establishes no PETW-specific wind-speed quantity, return level, design wind, design load, Path C control, or official guidance.

1 Introduction and Literature Review

1.1 The response-variable problem in thunderstorm-wind extreme-value analysis

Extreme wind-speed analysis requires an unambiguous definition of the response variable before any statistical model is fitted. A single storm can simultaneously produce a 1 s instantaneous peak, a 3 s block-averaged gust, a 3 s moving-average maximum, a legacy instrument-response product, a minute-scale mean wind speed, an hourly maximum gust, and an hourly mean wind speed. When the spatial support differs, further distinctions arise among station peaks, event-level network peaks, reanalysis gust diagnostics, radar or dual-Doppler reconstructed winds, and simulated wind speeds. Although these quantities share the same physical unit, they are jointly defined by different peak-search methods, sampling rates, instrument response characteristics, measurement heights, exposure conditions, and spatial supports (Kun and Ahsan, 2014; Cook, 2023b; Lombardo and Zickar, 2019). A common unit is therefore not a sufficient condition for numerical comparability, and samples that do not distinguish among these endpoints cannot enter the same tail fit or conversion chain.

The nominally identical "3 s gust" may originate from different data products. The U.S. ASOS gust report was changed from a non-overlapping block average to a moving average, and legacy cup anemometers paired with strip-chart recorders possess dynamic response characteristics that differ from those of automated-station algorithms (Kun and Ahsan, 2014; Lombardo and Zickar, 2019; Krupar et al., 2016). The replacement of the Australian Dines pressure-tube anemometer by cup-based systems similarly altered the response properties of long-term records (Cechet and Sanabria, 2010). These findings demonstrate that when merging records across studies, instruments, or time periods, the original algorithm and measurement chain must be verified before specifying the origin, destination, and applicability conditions of any conversion; product labels alone do not establish equivalence.

1.2 The complete inference chain from wind-speed observation to design interface

The path from raw observation to engineering interface is not a single unit conversion but an ordered inference chain. The measurement chain and averaging time determine the statisti-

cal identity of the recorded quantity (Kun and Ahsan, 2014; Choi and Hidayat, 2002). Measurement height, exposure, roughness, and directional sector define the standardization endpoint (Simiu et al., 2007; Giammanco et al., 2016). Mechanism classification then assigns each recorded event to a synoptic, thunderstorm-outflow, tropical-cyclone, or other category (Gomes and Vickery, 1978; Cook et al., 2003; Twisdale and Vickery, 1992). Quality control and the observation process determine whether a potential event can be recorded and retained (Cook, 2023b; Spassiani et al., 2023). Event independence and declustering rules govern whether the resulting extreme-value sample is valid (Cook, 1982; Brabson and Palutikof, 2000). Only then can annual-maximum, peaks-over-threshold, and component models enter a candidate comparison under their respective assumptions (Palutikof et al., 1999; Harris, 2009). Sampling, model-form, measurement-and-conversion, membership, detection-process, and scenario uncertainties pervade every link (Lombardo and Ayyub, 2017; Cook, 2004). Figure 4 accordingly separates the conditional research interfaces that the literature can support from the design interfaces that still require additional approval.

1.3 Principal gaps in existing research

At the measurement-standardization level, systematic discrepancies persist between the nominal "gust" label and a traceable averaging algorithm. Historical instrument response, sampling-rate changes, and recorder replacements render peak products from different periods at the same station directly incomparable (Cechet and Sanabria, 2010; Kun and Ahsan, 2014). At the mechanism-classification and event-independence level, classification methods, event occurrence rates, and detection processes are frequently treated separately in the literature: some studies define mechanism labels but do not establish corresponding independence rules (Vallis et al., 2019; De Gaetano et al., 2014), while others perform extreme-value declustering without distinguishing among mechanisms (Brabson and Palutikof, 2000). At the spatial-support level, wind-speed endpoints from different platforms (surface stations, towers, radars, reanalyses, and simulations) possess incompatible temporal resolutions, spatial resolutions, and height supports, which restricts the direct merger of cross-study results (Mehner, 2020; Steinheuer and Friederichs, 2020; Krupar et al., 2016). At the uncertainty-propagation level, existing studies typically report only a subset of uncertainty sources (such as parameter-estimation intervals) while neglecting the joint contribution of measurement-and-conversion, membership, detection-process, and scenario uncertainties (Mohr et al., 2017; Lombardo and Ayyub, 2017; Cook, 2004). Local conversion factors and single-station return levels lack a valid transfer path to PETW conditions.

1.4 Scope and organization of this study

This study synthesizes definitions, methods, applicability conditions, and limitations from identity-matched literature, organized by scientific question and inference chain. The discussion covers wind-speed quantity definitions and standardization, mechanism-aware event populations, observation processes, candidate extreme-value analysis routes, six uncertainty categories, and the conditional synthesis of mixed-wind climates, followed by the interface conditions between EVA and engineering use that remain to be satisfied. This study does not establish a PETW-specific wind-speed quantity or return level, does not produce a design wind, design load, Path C decision, or official guidance; the design interface still requires separate approval by PI

The existing literature confirms that extreme-value analysis of thunderstorm winds and mixed-wind climates has not yet established a unified comparable quantity object across all analysis links. This conclusion constitutes the logical premise for entering Section 2: before any extreme-value method is discussed, the definition space and standardization conditions of the wind-speed quantity must first be delimited.

2 Wind-Speed Quantity Definitions and the Standardization Contract

2.1 Peak duration, averaging time, and gust nomenclature

The fundamental prerequisite for wind-speed extreme-value analysis is an unambiguous definition of the response variable. Different peak durations and averaging algorithms produce different statistical objects. The 1 s instantaneous peak reflects the shortest-timescale wind-speed fluctuation (De Gaetano et al., 2014). The 3 s averaged gust can be either a non-overlapping block average or an overlapping moving average, and the maximum-value statistics of the two algorithms differ (Lombardo and Zickar, 2019). The 3 s moving-average maximum is the WMO-recommended standard definition (Steinheuer and Friederichs, 2020), but not every "3 s gust" label in historical records corresponds to this definition. The approximately 5 s response product derives from the equivalent filtering characteristics of early instrument chains (Kun and Ahsan, 2014; Lombardo and Zickar, 2019). The 2 min mean wind speed is used in sustained-wind reports from some automated stations (Chen and Lombardo, 2020). The 5 min event-mean wind speed serves as a candidate reference in thunderstorm-profile research because thunderstorms are highly localized in space and time and a 10 min average may excessively smooth peak features (Choi, 2004). The 10 min mean wind speed is the reference quantity in most meteorological standards and some design codes (Hong and Ye, 2014). The 30 s slowly varying mean is used to separate the slowly varying component in nonstationary thunderstorm signals (Zhang et al., 2018). Hourly maximum gusts and hourly mean wind speeds are basic products of routine climate datasets (Zhang and Chen, 2015; Steinheuer and Friederichs, 2020). Treating the "3 s" label as a unified target variable ignores the fact that different measurement chains can produce different numerical values for the same event.

2.2 Moving average, block average, and instrument response

The overlapping moving average and the non-overlapping block average are not statistically equivalent. Given the same high-frequency wind-speed time series, the moving-average maximum is generally higher than the block-average maximum (Lombardo and Zickar, 2019). The instrument response function further modifies the recorded peak characteristics. The distance constant of a cup anemometer determines the degree of low-pass filtering applied to the wind-speed signal (Kun and Ahsan, 2014). The strip-chart recorder has a response time of approximately 0.2 s, which together with the cup anemometer time constant of approximately 0.3 s forms a first-order low-pass chain whose equivalent moving-average window is approximately 1 s (Kun and Ahsan, 2014). Ultrasonic anemometers typically sample at 1 Hz to 50 Hz, with a frequency-response upper limit far exceeding that of cup-based systems (Choi, 2004; De Gaetano et al., 2014). The "equivalent duration" reported in any source can therefore only be understood as a conclusion under that specific instrument chain and cannot serve directly as a universal conversion coefficient. The ASOS system's gust products changed from a 5 s block average to a 3 s moving average across different periods, illustrating product-change issues within a single observation network (Lombardo and Zickar, 2019; Krupar et al., 2016; Cook, 2023b).

2.3 Measurement height, exposure, roughness, and directional sector

The vertical distribution of wind speed makes measurement height a central element of the endpoint definition. The 10 m level commonly serves as the reference height, but field measurements may be taken near the surface, at various tower levels, or at remote-sensing observation heights (Choi, 2004; He et al., 2016). Height conversion requires explicit specification of origin, destination, roughness length, stability conditions, and the profile model (Simiu et al., 2007; Gi-

ammanco et al., 2016). Open terrain, urban areas, harbors, sea surfaces, hilltops, and complex terrain possess different exposure conditions and cannot be merged unconditionally (Giammanco et al., 2016; Li et al., 2021). Directional stratification further alters the effective roughness and the sample size within each sector (Cook, 1982; Lombardo and Ayyub, 2017; Zhang and Chen, 2015). Gust factors from hurricane studies are similarly constrained by land or sea exposure, measurement height, and boundary-layer assumptions and cannot serve as a universal conversion quantity for thunderstorm winds or PETW (Vickery PI and Skerlj PI 2005).

2.4 Vertical profiles, upper-air observations, and near-surface wind speed

The vertical structure of thunderstorm outflow differs from the commonly used neutral boundary-layer profile. Field studies have documented near-surface velocity enhancement and low-level-jet features, noting that the profile shape and the height of maximum wind speed vary with storm-relative position and surface roughness (Choi, 2004). Wind profilers, dual-Doppler radars, and LiDARs provide profile or volume observations above the standard surface-station height; their temporal averaging, spatial resolution, and lowest available height each differ (He et al., 2016; Mehner, 2020; Krupar et al., 2016). Extrapolation from upper-level winds to near-surface gusts therefore depends on event type, stability, roughness, averaging time, and standardization method (Mehner, 2020). Reanalysis gust diagnostics are gridded parameterized products and cannot be equated with point-station event peaks (Steinheuer and Friederichs, 2020). Upper-level peaks, layer-mean winds, and near-surface gusts must be treated as distinct propositions; source-specific profiles cannot be replaced by a universal power law.

2.5 The standardization field contract for PETW

Based on the preceding analysis, any wind-speed record entering PETW analysis must simultaneously report the following fields: original wind-speed quantity, peak window (1 s, 3 s, or other), averaging algorithm (moving average or block average), reference averaging time (such as 10 min or 1 h), sampling rate and instrument response characteristics, measurement height, exposure and roughness conditions, directional sector, spatial-support type (station peak, event-level network peak, reanalysis gust, or other), QC status, mechanism label, classification confidence, independence rule, conversion endpoint, and conversion uncertainty (Figure 1, Table 1). This field contract defines only the decision logic and minimum information requirements; it does not supply PETW-specific conversion formulae, coefficients, or standardization results. When any critical endpoint is missing, the corresponding record can enter only a sensitivity branch or internal log and cannot be used for tail estimation.

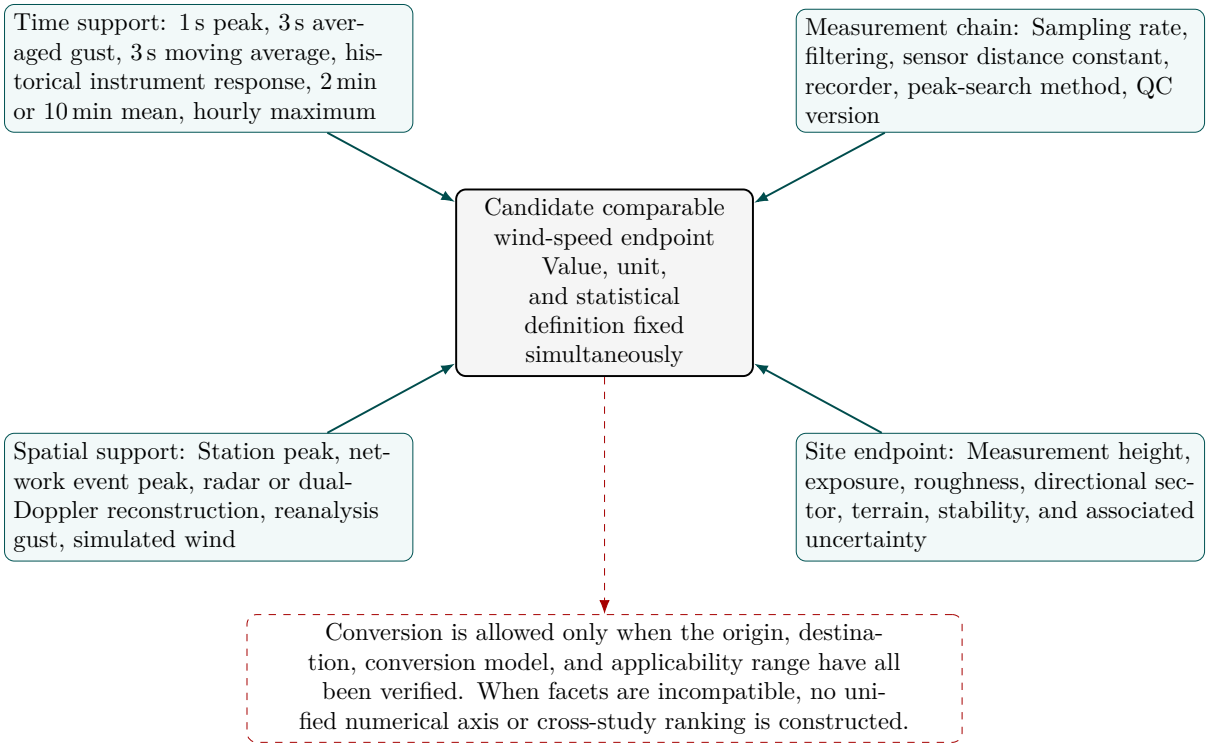


Figure 1: Wind-speed definitions and standardization comparison. Non-numerical facets represent time support, measurement chain, spatial support, and site endpoint; the facets do not form a unified numerical axis. Synthesized from the definitions, methods, and applicability boundaries of identity-matched literature; source-specific parameters are not transferred to PETW.

Table 1: Wind-speed variables and comparability conditions

Wind quantity	Peak algorithm	Time basis	Height	Exposure	Support	Comparison basis	Exclusion
1 s peak	Instantaneous or filtered peak	≤ 1 s	Source-dependent	Source-dependent	Station	Instrument response and filtering must be verified	Not equal to 3 s average
3 s averaged gust	Non-overlapping block average	3 s block	10 m or other	Must be specified	Station	Differs from moving average	Not interchangeable with moving average
3 s moving average max	WMO moving average	3 s moving window	Standard or actual	Must be specified	Station	WMO recommended definition	Not equal to block average
Approx. 5 s historical response	Instrument-equivalent filtering	$\sim$5 s	Instrument-dependent	Station-dependent	Station	Valid only under source instrument chain	Cannot be treated directly as 3 s
10 min mean	Fixed block	10 min	Usually 10 m	Specified exposure	Station	Standard meteorological reference	Cannot be merged with gust
Hourly max gust	Maximum gust within 1 h	Product-dependent	Source-dependent	Source-dependent	Station	Gust window must be specified	Not equal to annual maximum
Station peak	Single-station maximum	Definition-dependent	Source-dependent	Source-dependent	Single station	Complete definition chain required	Not equal to network peak
Event-network peak	Multi-station event maximum	Definition-dependent	Source-dependent	Source-dependent	Network	Deduplication and coverage required	Not equal to regional value
Reanalysis gust	Diagnostic parameterization	Gridded	Model level	Model	Grid	Not a point-station event-level gust	Not equal to observed gust
Simulated wind	Numerical simulation	Model-dependent	Model-dependent	Model-dependent	Simulation domain	Verification and bias correction required	Cannot be used directly for EVA

3 A Mechanism-Aware Wind-Event Population

3.1 Components and generating processes of mixed-wind climates

A mixed-wind climate is constituted by extreme wind events from multiple weather processes. Synoptic winds arise from large-scale pressure gradients and typically persist for hours to days over spatial scales of hundreds to thousands of kilometers (Gomes and Vickery, 1978; Cook et al., 2003). Extreme winds driven by tropical cyclones exhibit distinctive boundary-layer structures and spatial-field characteristics (He et al., 2016; Vickery PI and Skerlj PI 2005). Thunderstorm outflows (including downbursts and gust fronts) are highly localized in space and time, with durations as short as a few minutes (Choi, 2004; Choi and Hidayat, 2002). Tornadoes constitute yet another independent mechanism whose wind-field structure, spatial scale, and risk model differ fundamentally from those of conventional thunderstorm outflow (Fan and Pang, 2019; Lawrence A. Twisdale et al., 2023). The background synoptic system, as the carrying environment for convective events, can superpose on thunderstorm outflow (Choi, 2004). Each component may possess a different event occurrence rate, duration, spatial extent, and conditional tail distribution (Gomes and Vickery, 1978; Cook et al., 2003). At a given station, different mechanisms can produce different annual-maximum distributions, and the controlling mechanism can change with return period (Gomes and Vickery, 1978).

3.2 Thunderstorms, convective gusts, gust fronts, and non-thunderstorm events

Definitions of "thunderstorm wind" differ across regions and studies. Some studies rely on thunder or lightning labels to classify events (Choi and Hidayat, 2002; Choi and Tanurdjaja, 2002), while others use morphological features of the wind-speed time series (such as abrupt onset, nonstationarity, and anomalous gust factors) for classification (De Gaetano et al., 2014; Vallis et al., 2019). The gust front, as the leading-edge interface structure of thunderstorm outflow, may exhibit wind-speed characteristics different from those of the downburst core (Zhang et al., 2018). Intermediate and unresolved events are explicitly identified in multiple studies as records that cannot be silently assigned to any definite mechanism category (De Gaetano et al., 2014; Zhang et al., 2018). The same label (such as "thunderstorm") used in different regions or algorithms may correspond to different physical definitions and identification rules and cannot be treated as synonymous (Cook, 2024; Twisdale and Vickery, 1992).

3.3 Observational information and algorithm tiers for event classification

Event classification methods can be stratified by information source and degree of automation. Manual interpretation combining weather records, thunder reports, and surface observations represents the earliest classification approach (Twisdale and Vickery, 1992; Choi and Hidayat, 2002). Classification based on lightning and precipitation information draws on auxiliary data from meteorological detection networks (Spassiani et al., 2023; Mohr et al., 2017). Meteorological-variable rules (such as sudden temperature drops, pressure jumps, and wind-direction shifts) provide time-series-level classification criteria (Vallis et al., 2019; De Gaetano et al., 2014). Wind-speed morphological rules exploit anomalous features in gust factors, nonstationarity, and statistical moments (De Gaetano et al., 2014; Zhang et al., 2018). Automated classification algorithms and machine-learning classifiers (such as one-dimensional convolutional neural networks and self-organizing maps) can process large-scale data, but their performance metrics (accuracy, precision, and recall) describe label-discrimination performance and do not directly indicate physical missed-detection rates (Chen and Lombardo, 2020; Spassiani et al., 2023; Cook, 2024). A classifier's softmax output, accuracy score, and activation weights are not physical measures of detection probability (Chen and Lombardo, 2020).

3.4 Event independence, classification time windows, and spatial duplication

Event independence is a fundamental assumption of extreme-value analysis. The classification evidence window serves to gather mechanism information, while deadtime and declustering serve to establish extreme-value sample independence; these two time constants have different physical meanings and statistical purposes (Palutikof et al., 1999; Brabson and Palutikof, 2000). Independence intervals used in the literature vary across studies: Cook uses an approximately 2-day gust separation (Cook, 1982); Lombardo and Ayyub use 4-day and 0.5-day minimum time blocks for non-thunderstorm and thunderstorm events respectively (Lombardo and Ayyub, 2017); Choi and Tanurdjaja use 4 days and 24 h for large-scale and small-scale events respectively (Choi and Tanurdjaja, 2002); De Gaetano et al. propose preliminary separations of 72 h for extratropical lows, 4 h for thunderstorms, and 24 h for gust fronts (De Gaetano et al., 2014). Deadtime scanning in peaks-over-threshold analysis (such as 3 h to 96 h) is diagnosed via interval distributions and the independent-event index (Brabson and Palutikof, 2000). Spatial duplication of the same event across stations requires additional deduplication (Spassiani et al., 2023). All of the above time constants are valid only under their source conditions and cannot be elevated to universal defaults.

3.5 Event rate, classification probability, detection probability, and conditional amplitude

Extreme-value analysis of mixed-wind climates requires explicit distinction among four statistical objects. Event occurrence rate describes the frequency of a specific event type per unit time and depends on event definition, observation period, and independence rules (Cook et al., 2003; Twisdale and Vickery, 1992). Classification probability describes the mechanism assignment of a recorded event and depends on the classification algorithm and available information (Cook, 2024; Vallis et al., 2019). Detection probability describes the conditional probability that a potential event is recorded and retained, given station-network geometry, sensor status, reporting cadence, and QC rules (Spassiani et al., 2023). The conditional peak-amplitude distribution describes the peak statistical characteristics given a specific mechanism and specific observation conditions (Cook et al., 2003; Zhang et al., 2018). Conflation of these four objects introduces systematic bias into tail estimates. Unresolved events must be retained as at least three sensitivity scenarios: independent retention, conditional merging, and exclusion (Zhang et al., 2018; De Gaetano et al., 2014). Table 2 summarizes identification information, classification-probability semantics, independence rules, event-rate objects, detection processes, and unresolved-event handling for each mechanism category.

4 Observation Processes, Quality Control, and Detectability

4.1 Sensors, sampling rates, and station exposure

Different sensor types respond differently to the same event. Cup anemometers are constrained by distance constant and inertia; ultrasonic anemometers possess a higher frequency response (Kun and Ahsan, 2014; Choi, 2004). The ASOS system used different sensors and reporting rules in different periods (Cook, 2023b; Lombardo and Zickar, 2019). Tower observations can provide multi-height synchronous records (Choi, 2004), but measurement heights range from 2.25 m (such as StickNet) to several hundred meters (Mehner, 2020; Giammanco et al., 2016). LiDAR and radar observations supply remote-sensing profiles (He et al., 2016; Mehner, 2020) whose spatial and temporal resolutions differ from those of surface observations. Reanalysis diagnostic gusts are parameterized estimates (Steinheuer and Friederichs, 2020). The

same event may produce different peak values and different missing-data states across different measurement chains.

4.2 Missing data, peak deletion, outliers, and QC versions

The treatment of missing data and anomalies in observational records directly alters the extreme-value sample. The real-time quality control of the U.S. ASOS formerly suppressed certain peaks based on conditional relationships among the short-duration monitor value, the 2 min mean wind speed, and the 3 s gust; after rule updates, deleted peaks still cannot be uniquely recovered from the retained record (Cook, 2023b). Physical-range screening, despiking, and precipitation effects may also remove genuine events or retain nonphysical anomalies (Cook, 2024; Huang and Gu, 2019). The QC version, station-exclusion rationale, and unrecoverable data should be documented alongside the final sample; screening thresholds adopted in single-station studies or specific POT analyses are applicable only to their respective datasets and diagnostics and cannot become universal rules (Brabson and Palutikof, 2000; Mohr et al., 2017).

4.3 Instrument bias and measurement-chain uncertainty

Instrument replacement is a principal source of inhomogeneity in long-term wind-speed records. The replacement of the Dines pressure-tube anemometer by cup anemometers in Australian Region A produced systematic differences in extreme-value estimates (Cechet and Sanabria, 2010). At 17 of 31 stations, the 500-year return-level estimate from the replacement segment fell below the lower bound of the Dines-segment estimate (Cechet and Sanabria, 2010). Changes in filtering characteristics, distance constants, and recorder response times should all be tracked as sources of time-series inhomogeneity (Kun and Ahsan, 2014). Instrument bias cannot be corrected with a single unvalidated global factor.

4.4 Detectability, station density, and the potential-event process

The potential physical-event process and the observational recording process must be separated. Detection probability is the conditional probability that an event is recorded and retained, given the potential event, station-network geometry, sensor status, reporting cadence, and quality-control conditions. An Australian convective-gust climatology explicitly described the effect of station-network density on the detection process through a Bayesian hierarchical model within its observation period, event threshold, and automated-station spatial-aggregation conditions (Spassiani et al., 2023). That result demonstrates that the detection model must retain network and data conditions and does not provide a portable universal probability. Where a source has not established an observation model, the analysis can only record station-network coverage, missing-data patterns, QC deletions, and reporting intervals as detectability factors and cannot assign a numerical detection probability.

4.5 Multi-platform and spatiotemporal support

Station peaks, event-level network peaks, radar reconstructions, and reanalysis diagnostics possess different spatial and temporal supports (Mehner, 2020; Steinheuer and Friederichs, 2020; Krupar et al., 2016). Dual-Doppler data represent a reconstructed field above a finite spatial resolution and a minimum analysis height; reanalysis gusts are diagnostic quantities on a coarser grid. Neither can become a surface single-station gust through unit conversion alone. Different platforms can enter the same extreme-value sample only when their temporal averaging, spatial support, measurement height, and event-mechanism endpoints are compatible.

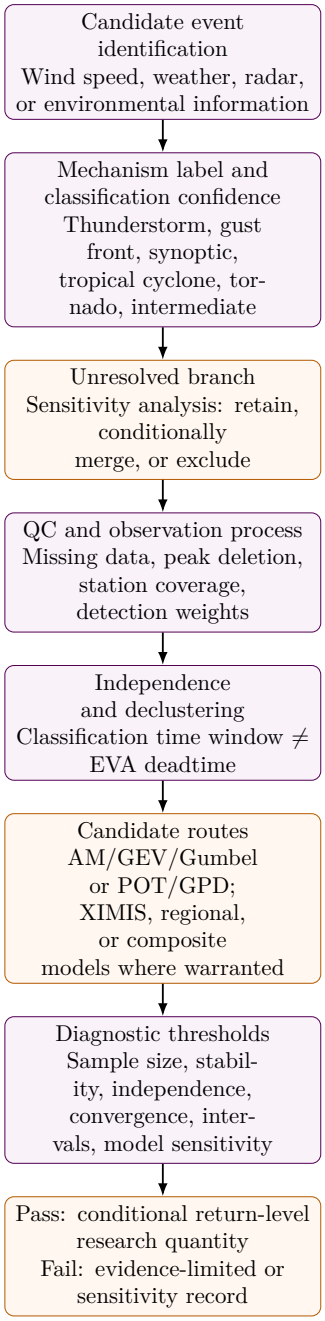


Figure 2: Methodological flow from event identification to EVA. Mechanism evidence and classification, observation and quality control, unresolved-event branches, cross-station deduplication, event independence, and candidate EVA diagnostics are connected in sequence. The classification evidence window differs from the declustering deadtime, and classification probability differs from detection probability. Synthesized from the definitions, methods, and applicability boundaries of identity-matched literature; source-specific parameters are not transferred to PETW.

5 Extreme-Value Analysis Routes and Event Independence

5.1 Annual maxima, GEV, and Gumbel

The annual-maximum method forms the sample from one extreme per year and fits a Generalized Extreme Value (GEV) distribution. The Gumbel distribution (GEV shape parameter $\xi = 0$) is a special case of the GEV (Palutikof et al., 1999). The block definition (calendar year or climate year), effective number of extremes, and information-use efficiency of the annual-maximum method depend on record length and mechanism structure (Hong and Ye, 2014; Cook et al., 2003). When a wind-speed series contains multiple mechanisms, annual maxima may originate from different categories, producing a mixed tail (Gomes and Vickery, 1978; Twisdale and Vickery, 1992). Under certain climate conditions, a Gumbel fit to thunderstorm annual maxima has been reported as conservative (Lombardo and Zickar, 2019), but distribution choices in the source do not constitute a universally applicable recommendation. The finite upper bound of the GEV (when the shape parameter is negative) may produce implausibly low values at long return periods (Hong and Ye, 2014). In regional frequency analysis, some clusters pass the Gumbel goodness-of-fit test while the GEV upper bound is unreasonable; the choice between the two distributions should be based on diagnostics rather than defaults (Hong and Ye, 2014).

5.2 POT, GPD, threshold, and independence diagnostics

The peaks-over-threshold (POT) method combined with the Generalized Pareto Distribution (GPD) exploits more extreme-value information (Palutikof et al., 1999; Brabson and Palutikof, 2000). Threshold selection must simultaneously consider the mean residual life plot, parameter stability, and exceedance rate (Brabson and Palutikof, 2000; Mohr et al., 2017). A high threshold leads to sample insufficiency; a low threshold introduces dependence and bias (Brabson and Palutikof, 2000). Deadtime and declustering rules ensure the independence of threshold exceedances (Brabson and Palutikof, 2000; Palutikof et al., 1999). The independent-event index ε is diagnosed via interval distributions and varies under different threshold-deadtime combinations (Brabson and Palutikof, 2000). POT/GPD becomes a candidate route only after the wind-speed quantity definition and event independence have been established. Source-specific thresholds, deadtimes, and return levels cannot serve as universal PETW parameters (Brabson and Palutikof, 2000; Mohr et al., 2017).

5.3 XIMIS, independent storms, and penultimate methods

XIMIS is a penultimate extreme-value method that starts from a finite event rate and parent distribution, treating independent storms, m-day maxima, thunderstorm maxima, and cyclone maxima (Harris, 2009). XIMIS exploits finite-rate information, left censoring, and rounding-perturbation removal, selecting a Type I transformation at large extrapolation (Harris, 2009; Cook, 2024). The Method of Independent Storms expands the annual-maximum sample by identifying independent storm events, defining storm boundaries via lulls (downward crossings below a threshold) (Cook, 1982). Penultimate methods address the asymptotic-convergence bias of finite samples using improved plotting positions and weighted least-squares estimation (Harris, 2009). These three approaches differ in assumptions, inputs, and statistical objectives: XIMIS emphasizes parent-process information and finite-rate correction, the Method of Independent Storms emphasizes event-level sample construction, and penultimate methods emphasize improved asymptotic approximation under finite samples (Harris, 2009; Cook, 1982). The three cannot be regarded as interchangeable implementations of the same method family.

5.4 Regional frequency, directionality, multivariate, and composite routes

Regional frequency analysis reduces quantile uncertainty from short records by pooling information across multiple stations (Hong and Ye, 2014). Discordancy and heterogeneity tests must be performed before pooling to verify regional homogeneity (Hong and Ye, 2014). Directional analysis stratifies extremes by directional sector and requires reporting of the sample size in each sector (Cook, 1982; Zhang and Chen, 2015). A multivariate framework (such as a Gaussian copula) can treat joint directional distributions (Zhang and Chen, 2015), but model selection and dependence-structure assumptions introduce additional model-form uncertainty. Multivariate and load-effect analyses can serve only as conditional descriptions of the EVA-to-design interface (Zhang and Chen, 2015).

5.5 Mechanism combination and occurrence-rate weighting

Component tails, event occurrence rates, dependence assumptions, and composite return levels must be modeled separately (Gomes and Vickery, 1978; Cook et al., 2003). Under an independent-mechanism assumption, the annual-maximum CDF of the mixed climate equals the product of the component CDFs (Gomes and Vickery, 1978). Before merging components, the separability of categories, the compatibility of wind-speed endpoints, and the independence assumptions must be stated (Cook et al., 2003; Cook, 2004). When a mechanism is masked or a rare-mechanism sample is insufficient, direct merging may underestimate long-return-period wind speeds (Cook, 2004; Gomes and Vickery, 1978). The composite expression is only a candidate research interface; without verified mechanism separation and tail diagnostics, it cannot serve as a final result (Cook, 2004).

6 Return Levels and Uncertainty Propagation

6.1 Sampling uncertainty

Record length, event count, number of annual maxima, and extrapolation distance jointly control sampling uncertainty. Under short records, POT/GPD parameter uncertainty grows with extrapolation distance (Brabson and Palutikof, 2000). Regional frequency analysis can mitigate single-station record insufficiency through multi-station information, but as return period increases the dispersion between regional and single-station quantiles may still widen (Hong and Ye, 2014). Comparisons based on synthetic parent distributions and simulated records further show that the estimation standard error and mean bias are distinct reliability metrics (Cook, 2023a). Source-specific numerical results are interpretable only when record construction, target return period, model, and interval definition are all retained; they cannot be transferred as PETW error bounds.

6.2 Model-form uncertainty

Threshold, block length, deadtime, distribution family, shape-parameter constraint, parent distribution, and stationarity assumption all belong to model-form choices (Palutikof et al., 1999; Cook, 2023a). Synthetic parent-distribution and simulated-record studies show that GEV and GPD biases vary with parent-distribution characteristics and extrapolation distance; this conclusion constitutes reliability evidence under the examined model conditions, not a route ranking for observed stations (Cook, 2023a). Mixed components may also produce an apparent shape parameter that differs from that of any single component. Gumbel and GEV can yield different long-return-period estimates under different record conditions (Hong and Ye, 2014; Lombardo and Zickar, 2019); this route discrepancy cannot be hidden within a single parameter standard error.

6.3 Measurement and conversion uncertainty

Averaging time, instrument response, height, exposure, roughness, direction, and platform conversion all introduce measurement and conversion uncertainty (Simiu et al., 2007; Giammanco et al., 2016; Cechet and Sanabria, 2010). Directional-sector stratification can express part of the roughness variation but cannot eliminate the uncertainty in roughness models and endpoint selections (Lombardo and Ayyub, 2017). Instrument replacements in Australian long-term records demonstrate that measurement-chain inhomogeneity can substantively alter extreme-value estimates (Cechet and Sanabria, 2010). Conversion errors cannot be assumed mutually independent, identically distributed, or independent of event mechanism by default.

6.4 Membership uncertainty

Classification errors assign events to the wrong mechanism category, altering the composition and conditional tail of category samples (Cook, 2024; Vallis et al., 2019). The attribution scheme for intermediate and unresolved events directly affects category occurrence rates and tail estimates (Zhang et al., 2018; De Gaetano et al., 2014). Classification performance metrics (accuracy, precision, and recall) describe label-discrimination ability and cannot substitute for detection probability (Chen and Lombardo, 2020). Long-duration convective events assigned to the synoptic ensemble may simultaneously alter both tails' long-term estimates (Vallis et al., 2019).

6.5 Detection-process uncertainty

Station-network density, reporting cadence, instrument status, missing data, and QC peak deletion determine whether potential events enter the record (Spassiani et al., 2023; Cook, 2023b). The spatial locality of thunderstorm outflow means that a sparse network may completely miss strong events (Aboshosha et al., 2020). QC peak deletion may remove valid extreme events (Cook, 2023b). Without an explicit detection model, only scenario or sensitivity analyses can be conducted; detection probability cannot be assigned a numerical value.

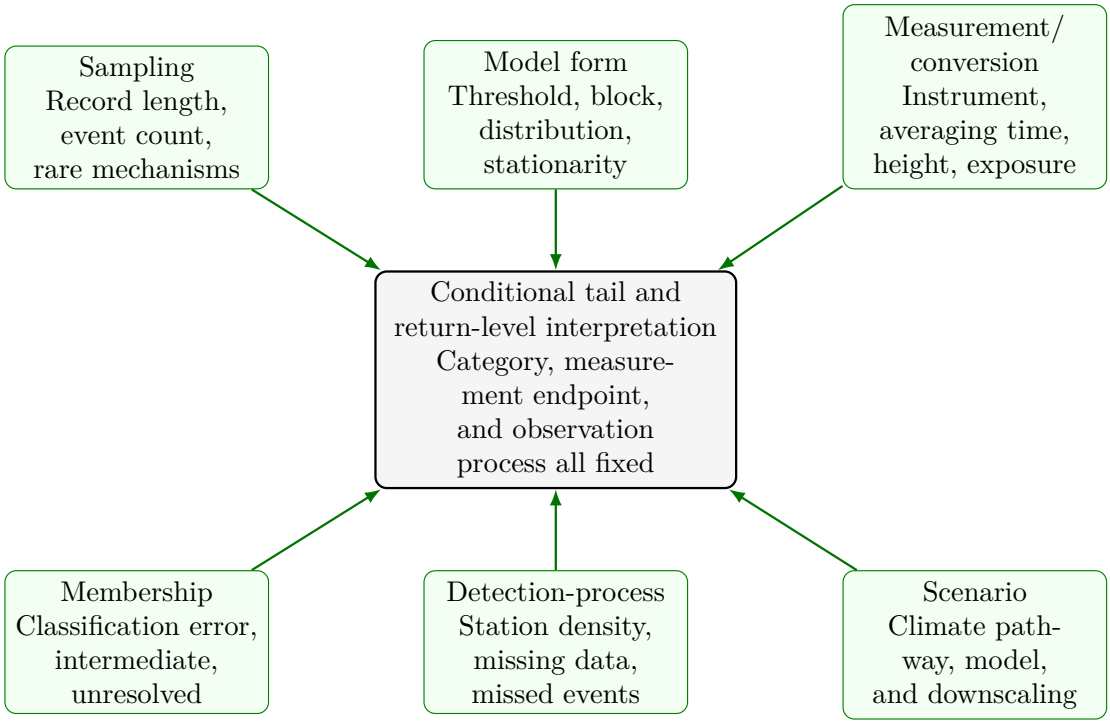
6.6 Scenario uncertainty and the reporting stopping rule

Historical versus future periods, global climate models (GCMs), downscaling methods, and convective proxy quantities belong to scenario uncertainty (Trapp et al., 2010; Wang et al., 2026; Wu et al., 2026). Changes in convective environmental proxies cannot be directly transcribed into future 3 s gust changes (Trapp et al., 2010; Wang et al., 2026). The confidence in frequency changes of tornadoes and thunderstorms under climate change has been assessed as "low" and "low" respectively (Wu et al., 2026). Reporting must retain all six uncertainty categories (Figure 3). When critical endpoints are unclear, diagnostics are unstable, samples are insufficient, or models do not converge, long-return-period extrapolation should be stopped; the upper bound of a sensitivity analysis should not be labeled a design value.

7 Conditional Synthesis of Mixed-Wind Climates and Thunderstorm Extremes

7.1 Conditional distributions precede unconditional mixture distributions

When analyzing extreme wind speeds in mixed-wind climates, conditioning on mechanism, measurement height, exposure, direction, measurement chain, and detection process should take priority (Cook et al., 2003; Zhang et al., 2018). An unconditional mixture distribution can be



Stopping condition: when a critical component cannot be identified, diagnostics are unstable, or the model does not converge, do not report ostensibly precise long-return-period values, and do not relabel an interval upper bound as a design value.

Figure 3: Propagation of six uncertainty categories into conditional tail interpretation. Sampling, model form, measurement/conversion, membership, detection-process, and scenario uncertainties are retained separately; membership is not merged with detection-process, and scenario is not subsumed under model form. Synthesized from the definitions, methods, and applicability boundaries of identity-matched literature; source-specific parameters are not transferred to PETW.

discussed only after category definitions, occurrence rates, wind-speed endpoints, and dependence assumptions have all been confirmed as compatible (Gomes and Vickery, 1978; Cook, 2004). Fitting a single distribution directly to an unclassified extreme-value sample may produce an apparent Type II (Fréchet) characteristic that is an artifact of the mixed tail rather than a reflection of the underlying physical process (Cook, 2004).

7.2 Component tails, occurrence rates, and composite return levels

The dominant mechanism at high return periods is not necessarily the dominant mechanism across all wind-speed ranges (Cook et al., 2003; Gomes and Vickery, 1978). Component occurrence rates and conditional amplitudes must be modeled and reported separately. Source-specific component analyses show that the high-wind-speed tail at Onslow is controlled by tropical cyclones while the corresponding tail at Brisbane is controlled by thunderstorms; these conclusions depend on the event classification, occurrence rates, and fitting construction at each location (Cook et al., 2003). U.S. station studies similarly show that the relative influence of thunderstorm and non-thunderstorm components can change with extrapolation range (Lombardo and Ayyub, 2017). Synthetic-record studies further demonstrate that when a rare component is underrepresented in finite samples, composite quantiles may exhibit systematic bias (Gomes and Vickery, 1978). These results support component separation and sensitivity analysis but cannot supply PETW occurrence rates, dominant mechanisms, or return levels.

7.3 Thunderstorm vertical structure and spatial locality

Thunderstorm downbursts can exhibit near-surface velocity enhancement and low-level jets (Choi, 2004). Dual-Doppler radars, wind profilers, and surface stations are inconsistent in measurement height, spatial support, and temporal averaging; therefore, the discrepancy between upper-level and surface wind speeds in the source literature cannot be simplified into a single vertical conversion (Mehner, 2020; Krupar et al., 2016; He et al., 2016). The spatial footprint of thunderstorm outflow may also be smaller than the station-network spacing, causing strong events to be absent from station historical records (Aboshosha et al., 2020). Studies based on historical records and event simulations support sensitivity analysis for spatial omission, but their long-return-period results cannot be transferred as PETW conclusions. A vertical structure from a single region or a single event likewise cannot form a universal profile rule.

7.4 Unresolved events and the sensitivity branch

Unresolved events in mixed-wind-climate analysis must be retained as an explicit sensitivity branch (Zhang et al., 2018; De Gaetano et al., 2014). When an independent unresolved category is retained, its contribution to the overall tail depends on sample size and amplitude distribution. Conditional merging (distributing unresolved events proportionally into existing categories) alters each category's occurrence rate and conditional tail. Excluding unresolved events reduces the effective sample size and may introduce selection bias. These three treatments can produce different category rates and tail estimates (Zhang et al., 2018). Without PETW-specific data diagnostics, no final scheme can be selected.

7.5 From literature methods to subsequent candidate research quantities

The subsequent research objects that literature methods can support must simultaneously fix averaging time and peak window, measurement height and exposure endpoint, mechanism classification and event-independence rules, while retaining traceable observation conditions and all six uncertainty categories. This study delimits only these necessary conditions; it does not

establish a PETW-specific wind-speed quantity or PETW return level, nor does it provide corresponding numerical values, intervals, or computational expressions.

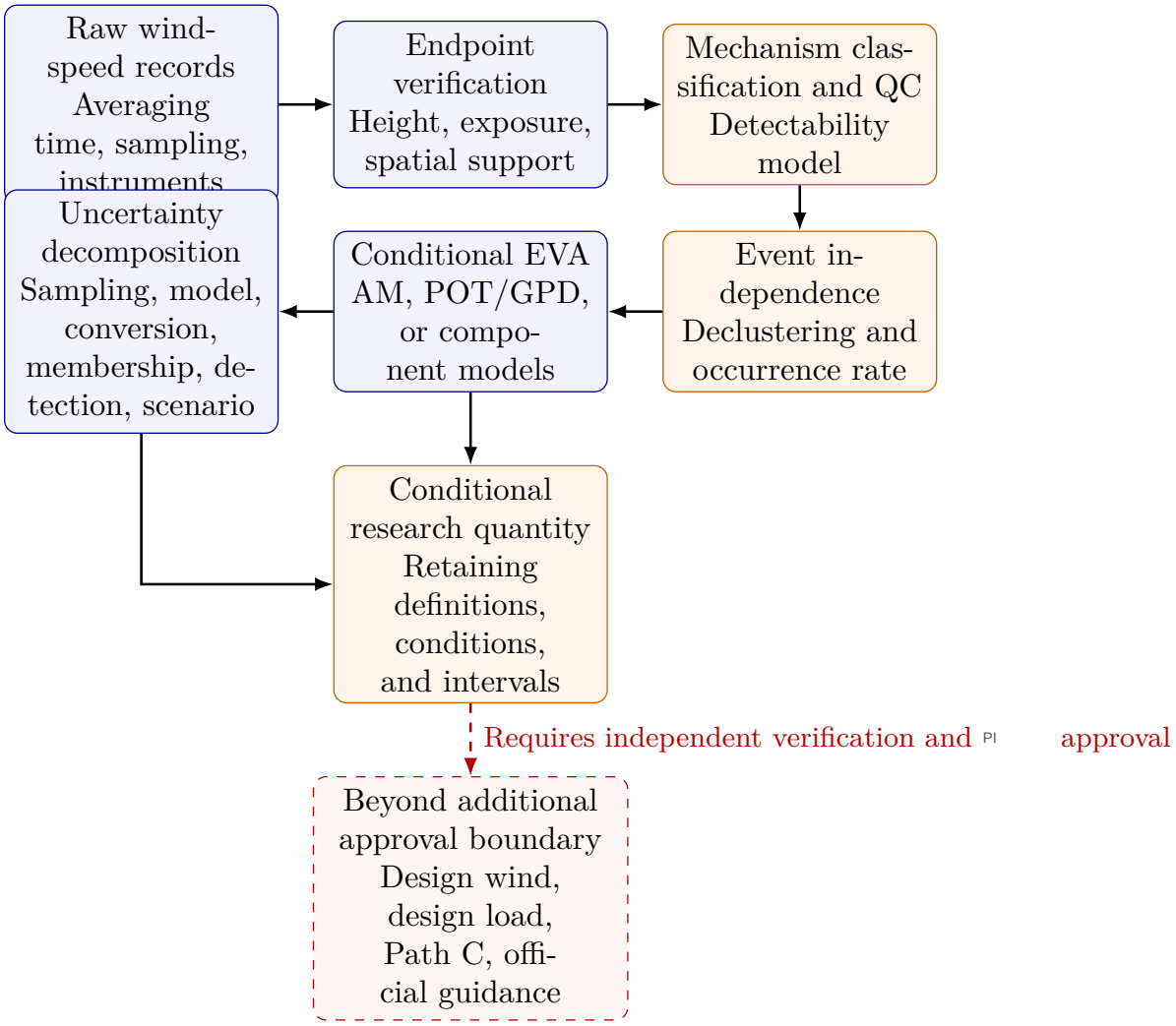


Figure 4: Evidence chain and PETW transfer boundary. Raw records pass through quantity definition, endpoint verification, mechanism classification, observation process, event independence, and conditional EVA before reaching the candidate research interface; design wind, design load, Path C, and official guidance lie beyond an additional approval boundary. Synthesized from the definitions, methods, and applicability boundaries of identity-matched literature; source-specific parameters are not transferred to PETW.

8 The Interface from EVA to Design Wind and Design Load

8.1 Basic wind speed, gust factor, and averaging-time consistency

The first step at the design interface is verifying the consistency among the EVA output quantity, the code quantity, and their averaging times, measurement chains, and statistical definitions. The ASCE 7 basic wind speed underwent a transition from fastest-mile to 3-s gust (Kun and Ahsan, 2014). A gust factor in the source literature is a conditional empirical quantity that depends on the peak definition of the numerator, the averaging period of the denominator, the exposure class, and the mechanism type (Choi and Hidayat, 2002; Giammanco et al., 2016; Vickery Peter and Skerlj 2005). Thunderstorm gust factors (3 s numerator / 10 min denominator)

are generally higher than the same-mechanism gust factors with a 1 h denominator (Choi and Hidayat, 2002; Mohr et al., 2017). Source-specific gust factors cannot directly become a PETW 3 s standardization factor.

8.2 Height, exposure, and terrain conversion endpoint conditions

Each height and exposure conversion must provide the origin, destination, effective height range, exposure class, roughness estimate, directional sector, applicable mechanism, and conversion uncertainty (Simiu et al., 2007; Giammanco et al., 2016; Li et al., 2021). The boundary-layer structures of hurricanes, thunderstorms, and synoptic systems differ; their height-conversion relationships are not interchangeable (Giammanco et al., 2016; Choi, 2004). Hurricane studies provide open-water-to-open-terrain conversions at 10 m height, for 1 min mean wind and peak 3 s gust, under specific roughness and profile assumptions (Simiu et al., 2007). These relationships are not applicable to thunderstorm winds. The vertical structure of tornado wind fields also differs from that of a conventional boundary-layer profile; the nominal 3 s engineering convention does not equate to measurement equivalence with an observed 3 s wind speed (Lawrence A. Twisdale et al., 2023).

8.3 Conditional return levels and the interpretation boundary of design quantities

Observed peaks, standardized research quantities, statistical return levels, design wind, and design load are quantities at different hierarchical levels (Wu et al., 2026). Local return levels, code comparisons, and single-station results cannot directly become PETW design inputs (Aboshosha et al., 2020; Li et al., 2021). The extrapolation reliability of long-return-period results (such as 700-year or 1000-year values) depends on record length, model selection, and the combined effect of all six uncertainty categories (Cook, 2023a; Aboshosha et al., 2020).

8.4 The approval boundary for Path C and official design guidance

The boundary shown in Figure 4 does not constitute a Path C decision; it does not approve a PETW design wind, compute a design load, modify a code, or issue official guidance. Any design-level use requires separate completion of data preparation, standardization verification, mechanism classification, observation-process modeling, tail diagnostics, and load-interface analysis, together with additional approval by PI

Table 2: Event mechanisms and population-construction conditions

Mechanism category	Identification information	Meaning of classification probability	Independence rule	Event-rate object	Detection process	Unresolved handling
Synoptic	Weather chart, pressure gradient, duration	Label assignment	Source-dependent (e.g., 2 days to 4 days)	Annual rate or block rate	Routine network	Not assigned by default
Tropical cyclone	Track, distance, pressure, center position	Label assignment	Independent-storm definition	Poisson rate	Track-to-station distance	Not assigned by default
Thunderstorm outflow	Thunder, lightning, wind-direction shift, gust factor	Label assignment	Source-dependent (e.g., 0.5 days to 4 h)	Annual rate	Spatial locality missed detection	Not assigned by default
Gust front	Gust-front morphology, nonstationary features	Label assignment	Event-level identification	Conditional rate	Requires nearby stations	Retain or analyze separately
Intermediate	Mixed features, long-duration convection	Physically unresolved	Sensitivity analysis required	Conditional rate	Requires auxiliary information	Retain as sensitivity branch
Tornado	Track, damage survey, EF rating	Independent classification	Independent model	Poisson rate	Depends on survey coverage	Independent branch
Unresolved	Cannot be assigned to any of the above	Insufficient information	Multi-scheme treatment required	Conditional rate	Information-limited	Independent retention, merging, or exclusion

9 A Minimum Analysis Framework for PETW EVA

9.1 Event-level data structure

The event-level data structure for PETW EVA must include at minimum the following fields: event identifier, original wind-speed quantity, peak window and averaging algorithm, sampling rate, measurement height, exposure and roughness conditions, instrument type and response characteristics, QC status and version, mechanism-category label, classification confidence, independence rule, detection-model status and its validated output, conversion endpoint, and the six uncertainty markers. Detection-model output can enter the analysis only when the model definition, training objective, and validation semantics are clear (Spassiani et al., 2023). Classification probability must not be entered in the detection-probability field.

9.2 Input eligibility gate

Before entering EVA, each event record must pass the following checks: wind-speed quantity identity confirmation, units and physical range, wind-speed definition and averaging time, measurement height and exposure, instrument type and calibration status, QC version and known deletions, information basis of the mechanism label, source and applicability conditions of the independence rule, and missing-data status and coverage. Records that do not pass the eligibility gate can enter only a sensitivity branch and do not participate in the primary tail estimate.

9.3 Candidate EVA comparison sequence

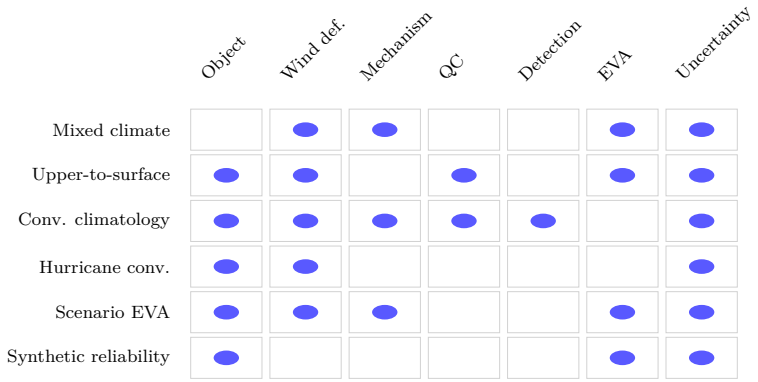
Candidate EVA comparison begins with quantity-endpoint consistency and event-level independent samples. Annual maxima with GEV or Gumbel and POT/GPD can be evaluated in parallel under their respective diagnostic conditions; XIMIS, independent storms, regional frequency, directionality, or composite routes enter the comparison only when the input information and model assumptions are satisfied. Each route must retain all six uncertainty categories, and conditional return levels should be interpreted only after diagnostics have stabilized. Figure 2, Figure 5, and Table 3 present this comparison framework but do not specify the final PETW route; the design interface remains subject to subsequent approval.

9.4 Uncertainty propagation and stopping conditions

The results of each candidate EVA route must itemize the source and magnitude of all six uncertainty categories (Table 3). Without evidence of joint dependence structure, the six interval values cannot be added, multiplied, or synthesized into a single total interval. When a critical quantity definition is incomplete (such as an unspecified peak window or averaging algorithm for the wind-speed quantity), event independence does not hold, the detection process cannot be accounted for, tail diagnostics are unstable, or the extrapolation distance is not credible, long-return-period extrapolation should be stopped (Cook, 2023a; Palutikof et al., 1999).

9.5 Interface with Step 2 and subsequent design analysis

The output of this study is a conditional methodological synthesis and an evidence-boundary record; its input is the definitions, methods, and applicability conditions of identity-matched sources. This study does not absorb Step 2 wind-field governance conclusions, does not replace load analysis, and does not produce a design wind, design load, or Path C decision. Interface conditions for subsequent PETW analysis include: a standardized event-level data structure, a verified mechanism-classification scheme, diagnosed independence rules, comparison results



• indicates the field is supported by identity-matched sources; blank indicates not established by that source category. The matrix identifies interface gaps; it does not rank study quality or imply a unified numerical scale.

Figure 5: Literature-method and evidence-field matrix. Rows represent methods or evidence categories; columns represent research object, definition endpoint, mechanism information, quality control, detection model, EVA route, and uncertainty coverage. The matrix indicates only whether the corresponding field has been explicitly addressed, not a study-quality ranking. Synthesized from the definitions, methods, and applicability boundaries of identity-matched literature; source-specific parameters are not transferred to PETW.

from at least two candidate EVA routes, itemized records of all six uncertainty categories, and explicit prohibited-transfer boundaries.

Table 3: EVA routes and uncertainty cross-reference

Route	Target and endpoint	Sample construction	Independence	Required diagnostics	Uncertainties represented	Stopping conditions	Transfer excluded
AM/GEV	Block maximum	Annual or seasonal block	Block definition	Shape diagnostics, Q–Q plot	Sampling, model form	When shape is unstable	Source block and shape
AM/Gumbel	Block maximum	Annual block	Block definition	Type I assumption	Sampling, model form	When heavy-tail evidence exists	Source results and intervals
POT/GPD	Threshold exceedance	Exceedance events	Deadtime declustering	Threshold stability, MRL	Sampling, model form, measurement	Unstable threshold	Source threshold and deadtime
XIMIS	Storm maximum	Lull separation	Event separation	Parent distribution, finite rate	Sampling, model form, measurement	Unreliable rate	Source w, rT, return values
Method of Independent Storms	Storm-level maximum	Lull separation	Inter-storm independence	Threshold sensitivity, run test	Sampling, model form	When interval criterion fails	Source threshold and parameters
Penultimate	Finite-sample correction	Same as XIMIS	Same as XIMIS	Asymptotic diagnostics	Sampling, model form	Insufficient convergence	Source corrections and parameters
Regional frequency	Regional quantile	Multi-station pooling	Station dependence	Homogeneity, discordancy	Sampling, model form, membership	High heterogeneity	Source clusters and results
Directionality	Directional quantile	Sector stratification	Sector definition	Sector sample size	Sampling, membership	When sector is too small	Source sectors and results
Composite model	Mixed return level	Component separation	Cross-component	Category completeness	All six categories	Inseparable components	Source rates and combinations

10 Study Limitations and Requirements for Further Work

10.1 Applicability boundaries of current scientific evidence

Wind-speed extreme-value analyses in the existing literature concentrate primarily on specific regions (North America, Europe, and Australia); climate conditions, measurement chains, and classification methods differ substantially across regions (Lombardo and Zickar, 2019; Mohr et al., 2017; Zhang et al., 2018). Short record lengths limit the reliability of long-return-period estimates (Brabson and Palutikof, 2000; Hong and Ye, 2014). Single-station case studies and local results lack a valid transfer path to PETW conditions (Aboshosha et al., 2020; Lombardo and Ayyub, 2017). Platform-endpoint inconsistencies across sources (such as differences in measurement height, exposure, averaging time, and gust definition) impede the direct merger of cross-study results (Cechet and Sanabria, 2010; Kun and Ahsan, 2014). Non-uniform mechanism definitions and classification methods render category rates and conditional tails from different studies directly incomparable (Vallis et al., 2019; De Gaetano et al., 2014).

10.2 Data and method limitations

Under short-record conditions, all EVA routes (annual maxima, POT/GPD, and component models) are constrained by sampling uncertainty (Palutikof et al., 1999; Cook, 2023a). The spatial locality of thunderstorm outflow means that a surface station network may completely miss strong events (Aboshosha et al., 2020). Insufficient vertical observations limit the reliability of conversion from upper-level wind speeds to near-surface gusts (Mehner, 2020). Instrument inhomogeneity introduces systematic bias in long-term records (Cechet and Sanabria, 2010). Classification uncertainty propagates through membership uncertainty into category-level tail estimates (Cook, 2024). Network sparsity propagates through detection-process uncertainty into event occurrence rates and sample completeness (Spassiani et al., 2023). The reliability of long-return-period extrapolation is jointly constrained by all six uncertainty categories.

10.3 PETW minimum next-step work package

Subsequent PETW analysis should first fix event-level fields and their definitions, then audit instrument response, measurement height, and exposure conditions to produce traceable station metadata. Mechanism labels need to be established concurrently with the sensitivity branch for unresolved events, while station-network coverage, missing-data patterns, and QC deletions are used to characterize the observation process. On this basis, AM/GEV, POT/GPD, and other conditionally eligible candidate routes can be compared on verified standardized samples, with itemized checks of all six uncertainty categories and convergence behavior. Whether to proceed to the design interface is to be decided by PI after this evidence has been assembled; this section makes no timeline commitment, provides no design value, and approves no route.

11 Concluding remarks

Wind-speed quantity definitions, the measurement chain, mechanism classification, the observation process, event independence, and EVA routes constitute a mutually constraining inference chain. The peak window, averaging algorithm, and instrument response determine the statistical identity of the recorded quantity; height and exposure endpoints delimit the scope of conversion. Mechanism classification shapes the component sample and its conditional tail; the observation process determines whether potential events can be recorded and retained. The classification evidence window cannot substitute for the declustering deadtime that establishes extreme-value

sample independence, and EVA routes must conform to the quantity endpoint, sample construction, and diagnostic conditions. Cross-study merging, adoption of uniform time constants, and long-return-period extrapolation can therefore proceed only after these conditions have been explicitly stated.

Conditional synthesis becomes feasible only after mechanism, measurement height, exposure, direction, measurement chain, and detection process have all been fixed, at which point component tails can be connected to composite return levels. Sampling, model-form, measurement/conversion, membership, detection-process, and scenario uncertainties act at different links and, in the absence of a joint dependence structure, cannot be compressed into a single total interval. This study has not established a PETW-specific wind-speed quantity or return level, nor has it produced a design wind, design load, Path C decision, or official design guidance. Any design-level use still requires independent data preparation, standardization verification, tail diagnostics, and additional approval by PI

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